LESSON 01 / 08

Counting is matching, construction is multiplication

To count a set X means to find $|X|$. Two master moves solve almost everything: **(A) construct** each object by a sequence of choices (multiply), **(B) match up** X bijectively with a set already counted (inherit its size). Every formula in this chapter is one of these, packaged.

KEY RESULT

Product rule and sum rule (with their exact hypotheses)

Product rule: a task in successive steps, step i offering n_i choices *regardless of earlier choices*, has $n_1 n_2 \cdots n_k$ outcomes. The independence clause is the hypothesis: "5 choices then 4 choices" for choosing 2 people from 5 counts **ordered** pairs (20), because $\{A, B\}$ is built twice.

Sum rule: if X is split into disjoint, exhaustive cases, $|X|$ is the sum of the case counts. Cases must be every case counted once, every object in some case.

KEY RESULT

Complement: negatives are rigid, positives are flexible

"At least one" / "no" / "not all" ⇒ count the complement and subtract from the total. Committees of 5 from 12 men and 8 women with at least one woman: $\binom{20}{5} - \binom{12}{5} = 15504 - 792 = 14712$, instead of five direct cases. Strings of length 4 over $\{0, 1, 2, 3, 4\}$ containing a 0: $5^4 - 4^4 = 625 - 256 = 369$.

WORKED EXAMPLE

Worked — cases where steps interact

4-digit numbers with distinct digits divisible by 5: split on the units digit (the thousands digit's options depend on it). Units 0: $9 \cdot 8 \cdot 7 = 504$. Units 5: $8 \cdot 8 \cdot 7 = 448$. Total 952. Splitting is the cure when independence fails in exactly one place.

INTUITION

Bijections you already own

Subsets of an n-set ↔ binary strings of length n (in/out per element): 2^n subsets. Increasing functions f with $f(1) < f(2) < f(3)$ into $\{1, \ldots, 5\}$ ↔ 3-subsets: $\binom{5}{3} = 10$. Pairs $A \subseteq B$ of subsets: each element has 3 states (in neither, in B only, in both): 3^n. Monotone/ordered-with-condition almost always means: find the unordered object it encodes.

COMMON TRAP

The three sins of counting (check every answer against them)

1. **Overcounting:** one object built by two choice paths. 2. **Undercounting:** cases not exhaustive. 3. **Broken independence:** multiplying counts that are not fixed per step. Cure for all three: verify a small case by listing. Also ask: objects labelled or unlabelled? order matters or not? is the answer a plausible size (never more than the total pool)?

ASSUMPTIONS & CONVENTIONS

Standing conventions

n, r are non-negative integers with $0 \le r \le n$ unless stated; $\binom{n}{r} = 0$ if $r < 0$ or $r > n$ (a useful convention in sums); $0! = 1$ (exactly one empty arrangement). "Arrangements" are ordered; "selections/committees" are unordered.

LESSON 02 / 08

Ordered selections

SHORT DERIVATION

nPr from filling labelled roles

Fill r labelled positions from n distinct objects: n options, then $n-1$, ..., then $n-r+1$ (each count fixed regardless of history). Hence

$$P(n,r)=\frac{n!}{(n-r)!}, \qquad P(n,n)=n!$$

Equivalently $P(n,r)$ counts injective functions from an r-set to an n-set; $n!$ counts bijections of an n-set. $0!=1$ is forced: there is exactly one empty arrangement.

KEY RESULT

Identical objects: label, count, divide by the symmetry

Arrangements of a multiset with $n_1,\dots,n_k$ copies of k types ($n=\sum n_i$):

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}$$

Proof: temporarily label all copies ($n!$ words); each unlabelled word is hit exactly $n_1!\cdots n_k!$ times (permute labels within each type). Division is legal ***iff every fibre has the same size*** — the exact-symmetry condition; where it fails (necklaces with identical beads), Part II's Burnside lemma is the fix.

Words from COMBINATORICS (C, O, I twice each, 13 letters): $\frac{13!}{2!\,2!\,2!}=778377600$. MATHEMATICS: $\frac{11!}{2!\,2!\,2!}=4989600$.

WORKED EXAMPLE

Worked — bundles inside identical-letter words

Words from DAAR with the two A's adjacent: bundle AA into one super-object; arrange $\{D, AA, R\}$: $3!=6$ (the identical A's give the bundle one internal order). Words: DAAR, AADR, ADAR, ADRA, ARDA, RADA ✓.

COMMON TRAP

Common traps — permutations

- Dividing by symmetry when the symmetry is not exact (some arrangements fixed by the swap).
- "Choose first, then second" silently counting ordered pairs for an unordered question — off by $r!$.
- Bundles: internal orders multiply only if the bundled objects are distinct; identical objects contribute 1.
- $P(n,r)$ with $r>n$ is 0, not a formula to evaluate.

LESSON 03 / 08

Around a table

KEY RESULT

Circular permutations: $(n-1)!$

n distinct people around a round table, seatings identified up to rotation: $(n-1)!$. Two proofs: anchor one person as reference and arrange the rest linearly; or divide $n!$ by the n rotations (every orbit has size exactly n for distinct people). If flips are also identified (necklaces of **distinct** beads, $n \geq 3$): $\frac{(n-1)!}{2}$.

Two specified people adjacent (round table): bundle, $2(n-2)!$; not adjacent: $(n-1)! - 2(n-2)! = (n-3)(n-2)!$. For 10 people with A, B non-adjacent: $9! - 2 \cdot 8! = 362880 - 80640 = 282240$.

The three restriction methods (linear arrangements)

KEY RESULT

Bundle, gap, relative order — pick by the wording

- "**Together/consecutive**": bundle the k objects into one super-object ($k!$ internal orders if distinct), arrange the $n-k+1$ objects.
- "**No two adjacent**": gap method. Arrange the m ordinary objects ($m!$), creating $m+1$ gaps; place the k special objects into **distinct** gaps: $\binom{m+1}{k}$ choices and $k!$ orders: total $m!\binom{m+1}{k}k!$. Six people with A, B, C pairwise non-adjacent: $3! \cdot \binom{4}{3} \cdot 3! = 144$.
- "**Fixed relative order**" **(A before B before C, not necessarily adjacent)**: among all $n!$ orders the k specified objects show each of their $k!$ relative orders equally often: answer $\frac{n!}{k!}$. "A before B" alone: $\frac{n!}{2}$.

WORKED EXAMPLE

Worked — composing methods

Six people P, Q, R, S, T, U in a row; P, Q together and R not at an end. Bundle PQ: $2 \cdot 5! = 240$ arrangements of the 5 objects. Subtract R at an end: 2 ends, then $4!$ arrangements of the rest, bundle order 2: $2 \cdot 4! \cdot 2 = 96$. Answer $240 - 96 = 144$. Order of operations: bundle first, then apply the positional restriction to the shrunken list.

Alternation: 5 men and 5 women alternating in a row: 2 patterns $\times 5! \cdot 5! = 28800$.

COMMON TRAP

Common traps — restrictions

- Gap method requires the ordinary objects arranged **first**, and needs $k \le m + 1$ to be possible at all.
- "Not adjacent" for a specified pair is bundle-and-subtract; "no two of a set adjacent" is the gap method. Swapping them is the classic error.
- Circular "not adjacent" via linear gap method: wrong geometry — use total minus bundled.
- Necklace division by 2 needs distinct beads (or Burnside); identical beads break exactness.

LESSON 04 / 08

Unordered selections

SHORT DERIVATION

Why dividing by r! is exact here

Every r-subset generates exactly $r!$ orderings, and every ordering comes from one subset: the ordered selections $P(n, r)$ split into equal fibres of size $r!$. So

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}, \qquad \binom{n}{r} = \binom{n}{n-r}$$

The symmetry is complementation: choosing what to keep = choosing what to discard. First question of every problem: do the objects come with positions/labels (permutations) or not (combinations)? Mixed problems do both, in that order.

KEY RESULT

Pascal and hockey-stick (proofs are one split each)

Pascal: $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ — split r-subsets by whether a fixed element is in (then choose $r-1$ more) or out (choose all r from the rest).

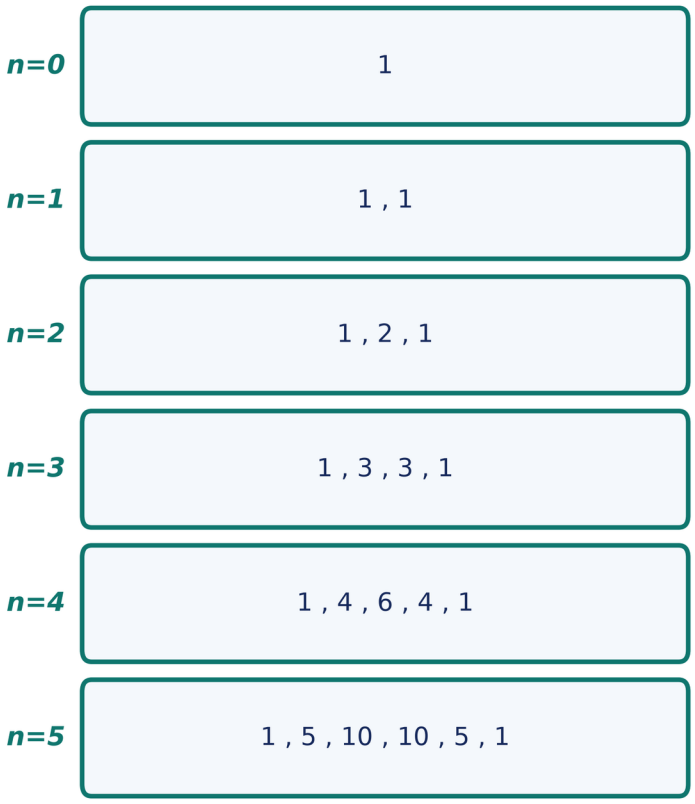
Hockey-stick: $\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$ — count $(r+1)$-subsets of $\{1, \ldots, n+1\}$ by their largest element $i+1$; the rest is an r-subset of $\{1, \ldots, i\}$. E.g. $\binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \binom{4}{1} = 10 = \binom{5}{2}$ (which is also $1+2+3+4$).

WORKED EXAMPLE

Worked — restricted selections

At least one of 4 special books among 5 chosen from 20: $\binom{20}{5} - \binom{16}{5} = 15504 - 4368 = 11136$ (complement again).

Team of 11 from 7 batsmen, 5 bowlers, 3 all-rounders, at least 3 batsmen and at least 2 bowlers: enumerate by all-rounders $a = 0..3$ and sum products $\binom{7}{b}\binom{5}{w}\binom{3}{a}$ over valid (b, w): $12 + 198 + 660 + 490 = 1360$. The template: one free variable plus bounds makes a finite table — check boundary triples twice, and **do not drop the $\binom{3}{a}$ factor** for the enumerated group (the classic table error: rows for $a = 1, 2$ are each 3 times their unweighted subsums).



Pascal's triangle — row n of $(a+b)^n$, each entry the sum of the two above

COMMON TRAP

Common traps — combinations

- $\binom{n}{r}$ computed for "arrange the chosen few" problems (or vice versa): decide order first.
- Case splits on "at least" that double-count overlaps — use complement or inclusion–exclusion (sheet 06).
- Boundary triples dropped in group-selection tables (the $(7, 3)$-type corner cases).
- Choosing from groups: multiply the independent group choices, then sum over the valid compositions — not the reverse.

LESSON 05 / 08

Identical objects into distinct boxes

SHORT DERIVATION

The stars-and-bars bijection

Distributing n **identical** balls into k **distinct** boxes is the same data as a count tuple $(x_1, \dots, x_k)$, $x_i \geq 0$, $\sum x_i = n$ — and the same as a word of n stars and $k-1$ bars (stars for box 1, bar, stars for box 2, ...). Choosing the bar positions among $n+k-1$ symbols:

$$\binom{n+k-1}{k-1} \quad \text{(empty boxes allowed)}$$

$$\binom{n-1}{k-1} \quad (\text{every box} \geq 1)$$

The positive case: pre-place one ball in each box, distribute the remaining $n-k$ freely. Non-negative solutions of $x+y+z=15$: $\binom{17}{2} = 136$; positive: $\binom{14}{2} = 91$ (difference $45 = 42 + 3$: exactly one zero $3 \cdot 14$, exactly two zeros 3 ✓).

KEY RESULT

The same formula in four costumes

- **Multisets:** choosing 12 fruits from 5 varieties (counts only): $\binom{12+5-1}{4} = \binom{16}{4} = 1820$.
- **Compositions:** ordered sums of n into k positive parts: $\binom{n-1}{k-1}$ (separators in the $n-1$ gaps between unit blocks); any number of parts: 2^{n-1}. Ordered sums of 10 into parts ≥ 2: shift $b_i = a_i - 1$, sum over the number of parts $k = 1..5$: $1 + 7 + 15 + 10 + 1 = 34$ (the $k=1$ term is the single-part sum $[10]$ itself — easy to drop).
- **Non-negative solutions with lower bounds:** shift each variable down to 0 first.
- **Upper bounds:** complement/inclusion–exclusion: $x_1 + x_2 + x_3 + x_4 = 20$, $x_1 \leq 5$: total $\binom{23}{3} = 1771$ minus ($x_1 \geq 6$, shift) $\binom{17}{3} = 680$ gives 1091.

COMMON TRAP

Common traps — stars and bars

- Identical vs distinct balls: distinct balls into boxes is k^n (functions), a completely different count. Read which objects are identical.
- Applying $\binom{n+k-1}{k-1}$ to ordered compositions with positive parts (that is $\binom{n-1}{k-1}$).
- Upper-bound problems with two violated bounds need genuine inclusion–exclusion, not one subtraction.
- Balls identical + boxes identical (partitions) has **no closed form** — Part II; do not invent one.

LESSON 06 / 08

The principle

SHORT DERIVATION

Inclusion–exclusion, proved by tracking one element

For bad events $A_1, \dots, A_n$ in a universe X:

$$\left|X \setminus \bigcup A_i\right| = \sum_{S \subseteq \{1..n\}} (-1)^{|S|} \left|\bigcap_{i \in S} A_i\right|$$

Proof: an element lying in exactly $t \geq 1$ of the sets is counted $\sum_{j=0}^{t} (-1)^j \binom{t}{j} = (1-1)^t = 0$ times; an element in none is counted once. **Universal recipe:** define bad events; compute the intersection size for a generic set of k specified events (usually $\binom{n}{k} \times$ something free); write the alternating sum.

Integers 1–1000 divisible by 2, 3 or 5:
$500 + 333 + 200 - 166 - 100 - 66 + 33 = 734$ (so divisible by none: 266).

KEY RESULT

Derangements: the flagship application

D_n = permutations of $\{1..n\}$ with **no** fixed point. With A_i = "i is fixed", a specified set of k fixed points leaves $(n-k)!$ free permutations:

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$$

$D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44, D_6 = 265$. Also $D_n = (n-1)(D_{n-1} + D_{n-2})$ (where does element 1 go? then whether its partner swaps back), and D_n is the **nearest integer to** $n!/e$ since the tail beyond n is $< \frac{1}{n+1} < \frac{1}{2}$.

Exactly k fixed points: $\binom{n}{k} D_{n-k}$ (for $n = 5$, exactly two correct letters: $\binom{5}{2} \cdot 2 = 20$; check: $44 + 20 + 45 + 10 + 0 + 1 = 120 = 5!$ ✓).

WORKED EXAMPLE

Worked — onto functions (IE in disguise)

Onto functions from a 5-set to a 3-set: total 3^5 minus those missing a value: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150$. Hands of 6 cards containing all four suits: $\binom{52}{6} - 4\binom{39}{6} + 6\binom{26}{6} - 4\binom{13}{6} = 8682544$.

COMMON TRAP

Common traps — inclusion–exclusion

- Sign alternation starting wrong: the empty intersection (whole universe) carries +.
- Intersection sizes computed for *exactly* k events instead of *at least these* k (IE wants the latter).
- Bounded stars-and-bars terms with negative top: $\binom{m}{k-1}$ for a negative-sum equation is **0** — terminate the sum, never evaluate.
- Derangement problems with only *some* positions forbidden: restrict the event set (then it is partial derangement / rook theory, Part II flavour).

LESSON 07 / 08

Existence by averaging

KEY RESULT

Pigeonhole principle (the art is the boxes)

$n+1$ objects in n boxes $\Rightarrow$ some box has ≥ 2. The theorem is trivial; every application is the choice of the box map so that "same box" gives the conclusion.

Odd-part box: from any $n+1$ integers in $\{1, \ldots, 2n\}$, two share the same odd part in $x = 2^a \cdot m$ (m odd) — only n odd parts exist — so one divides the other. Sharp: $\{n+1, \ldots, 2n\}$ has no such pair.

Geometry box: 5 points in a unit square: quarter it into 4 subsquares; two points share one, at distance $\leq \frac{\sqrt{2}}{2}$ (its diagonal).

Counting by last-step decomposition

KEY RESULT

Recurrences: split by the last block

If objects of size n end with one of a few block types, leaving a smaller instance of the same problem, the counts satisfy a recurrence. $2 \times n$ domino tilings: ends vertical (a_{n-1}) or two horizontal (a_{n-2}): $a_n = a_{n-1} + a_{n-2}$, $a_0 = a_1 = 1$, so $a_n = F_{n+1}$ (tilings of 2×10: 89). Binary strings with no consecutive 1s and subsets with no two consecutive give the same recurrence (F_{n+2}). Staircase with steps 1–2: $a_{10} = 89$; steps 1–3: tribonacci. Guard the base cases — a correct recurrence with wrong seeds is a wrong answer.

KEY RESULT

Recap — the chapter on one card

Product/sum/complement; bijection = same count. $P(n,r) = \frac{n!}{(n-r)!}$; identical objects $\frac{n!}{\prod n_i!}$; circular $(n-1)!$ (necklace $\frac{(n-1)!}{2}$ distinct beads). Bundle / gap $m! \binom{m+1}{k} k!$ / relative order $\frac{n!}{k!}$. $\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$; Pascal; hockey-stick. Stars and bars $\binom{n+k-1}{k-1}$ / positive $\binom{n-1}{k-1}$; compositions 2^{n-1}. IE alternating sum; $D_n = n! \sum \frac{(-1)^k}{k!} \approx n!/e$; onto $= \sum (-1)^j \binom{k}{j} (k-j)^n$. Pigeonhole: design the boxes. Recurrence: split by last block, mind the seeds.

COMMON TRAP

Pre-submission checklist

Order matters? Labels? Identical objects — is my division by symmetry exact (equal fibres)? Cases disjoint and exhaustive? Did "at least" get complemented? Bounds handled by shift/IE, with negative-top terms set to 0? Small case listed by hand? Answer size sane against the total pool?

INTUITION

Forward map

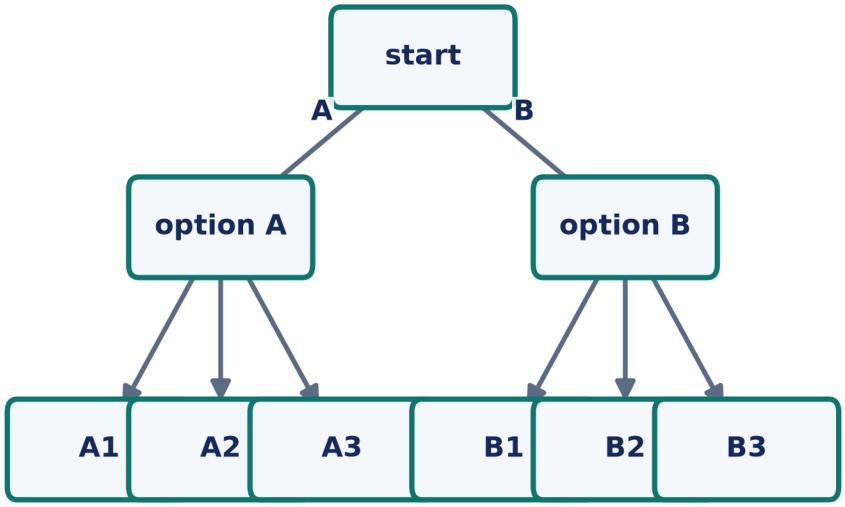
Binomial identities are this chapter counted twice (chapter 9); lattice paths and Catalan, Burnside–Pólya, partitions and Stirling numbers, intersecting families: Part II here. The through-line never changes: construct, split, or match up.

LESSON 08 / 08

Multiply along a path, add across disjoint paths

The product rule counts a sequence of independent decisions; the sum rule counts disjoint cases. A tree makes the distinction visible: each root-to-leaf path is one outcome, while the leaves are partitioned by their first decision.

A two-stage decision tree has 2 x 3 = 6 leaves



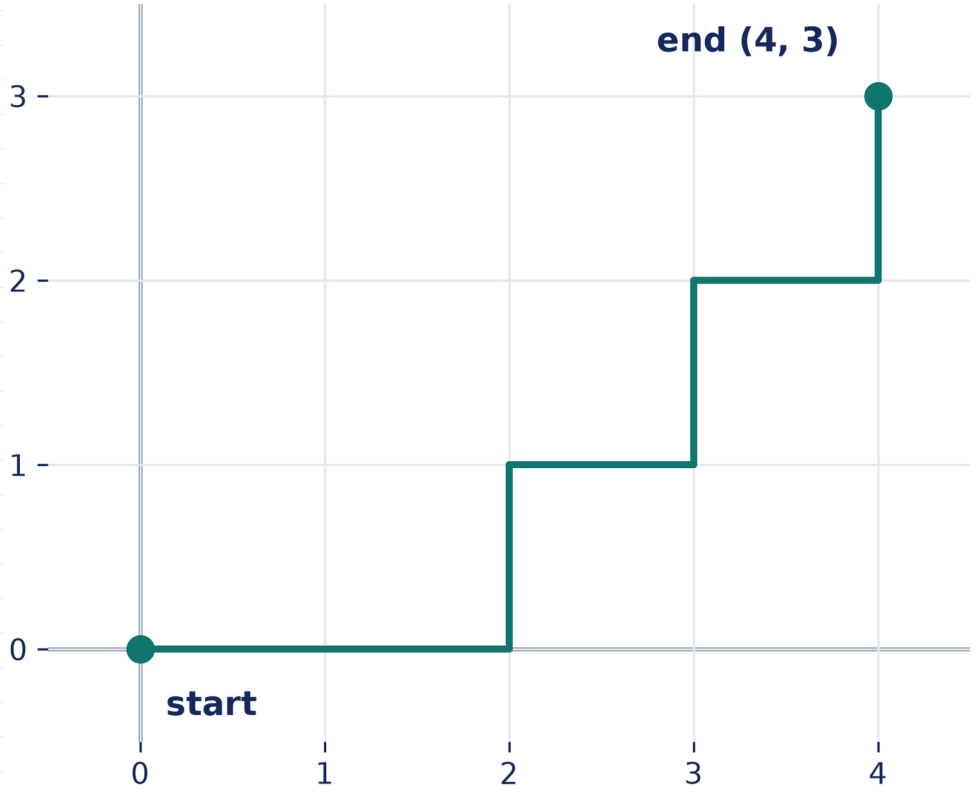
READ THE FIGURE
The first stage has two alternatives and the second has three for each alternative. There are six root-to-leaf paths, not 2 + 3 = 5.

If the second-stage number of choices changes with the first decision, do not multiply a single pair of numbers. Count each branch separately and then add the branch totals.

Lattice paths encode a word with identical letters

A shortest path from $(0,0)$ to $(4,3)$ using only right and up steps contains four R's and three U's. Every arrangement of the seven-letter word RRRRUUU gives exactly one path, so the count is $\binom{7}{3} = 35$.

One shortest path represented by R R U R U R U



READ THE FIGURE
Exactly seven unit steps occur. Choosing which three of the seven positions are U determines all four R positions, giving C(7,3) = C(7,4) = 35.

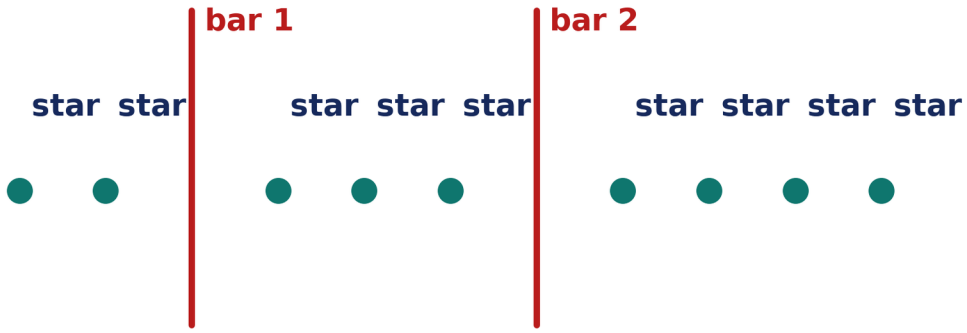
If diagonal or left/down steps are allowed, this binomial count no longer applies. The step restrictions are part of the model, not a minor detail.

Stars and bars count nonnegative distributions

The number of nonnegative integer solutions of $x_1 + x_2 + x_3 = 9$ is $\binom{9+3-1}{3-1} = \binom{11}{2} = 55$.

Arrange nine stars and two bars; the bars split the stars into three groups.

The arrangement ** | * | **** represents (2, 3, 4)**



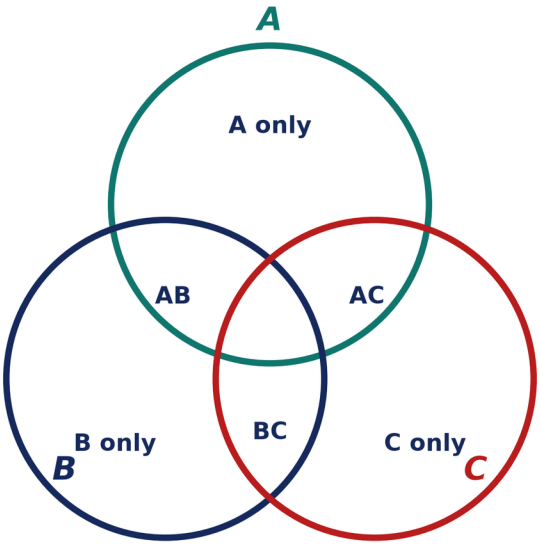
READ THE FIGURE

Nine stars and two dividers occupy eleven positions. Choosing the two bar positions determines the nonnegative group sizes. Empty groups are allowed.

For positive solutions, set $y_i = x_i - 1$ first. The transformed equation has a smaller right-hand side; forgetting that shift is the most common stars-and-bars error.

Inclusion–exclusion repairs double counting

For two sets, $|A \cup B| = |A| + |B| - |A \cap B|$. The intersection was counted once in $|A|$ and again in $|B|$, so subtract it once. For three sets, pairwise intersections are subtracted and the triple intersection is added back.



Three-set overlap regions must be counted with the correct multiplicity

READ THE FIGURE

The centre ABC occurs in all three set counts, all three pairwise counts and the triple count. Its net coefficient is 3 - 3 + 1 = 1.

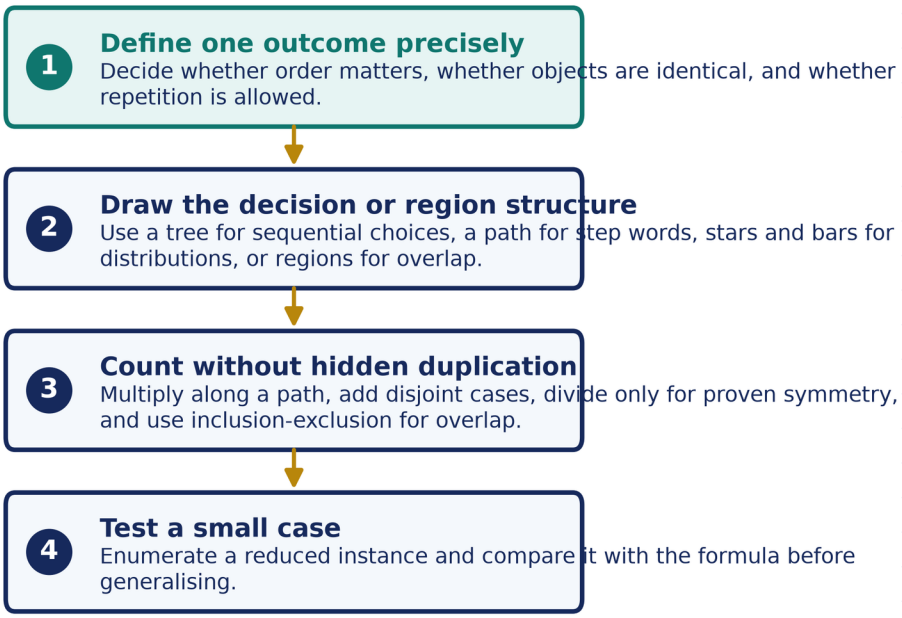
WORKED EXAMPLE

None of the bad events

If A_i is the event that condition i fails, then "no failure" has count $N - |A_1 \cup \cdots \cup A_m|$. Inclusion–exclusion is often used on the bad events and then subtracted from the total, rather than applied directly to the desired region.

A counting workflow

Choose the model before choosing a formula



READ THE FIGURE

Most counting errors are model errors: a correct permutation or combination formula applied to the wrong definition of an outcome.

KEY RESULT

Complement is a counting method

When "at least one" or "none" creates many cases, count the complementary event. For example, at least one success in n independent trials has probability $1-(1-p)^n$; the same complement idea works in ordinary enumeration.